BIF604: Comprehensive 200+ MCQs with Correct Answers

Section 1: Bioinformatics Tools (DAVID, STRING, Cytoscape)

DAVID (Database for Annotation, Visualization and Integrated Discovery)

1. **What does DAVID stand for?** Correct Answer: Database for Annotation, Visualization and Integrated Discovery
2. **What is the primary aim of DAVID?** Correct Answer: To provide functional interpretation of large gene datasets
3. **Is DAVID a free or licensed software?** Correct Answer: Free Bioinformatics Resource Software
4. **Which tool in DAVID is used for batch annotation?** Correct Answer: DAVID functional annotation Tool
5. **Which DAVID tool helps in identifying pathogens?** Correct Answer: DAVID NIAID Pathogen Genome Browser
6. **What is the significance of ranking genes in DAVID?** Correct Answer: Helps a quick focus on significant genes
7. **How many web-based functional annotation tool suites are embedded in DAVID?** Correct Answer: Five
8. **What does the DAVID gene ID conversion Tool do?** Correct Answer: Converts one type of gene ID to another
9. **Can DAVID visualize annotations?** Correct Answer: Yes
10. **What is the first step in the DAVID functional annotation workflow?** Correct Answer: Load Gene List
11. **What algorithm does DAVID use for gene-GO term enrichment?** Correct Answer: A novel algorithm to measure relationships among annotation terms
12. **Which report in DAVID provides a summary of results?** Correct Answer: Summary Page
13. **What is the purpose of the DAVID gene name viewer?** Correct Answer: To view names associated with genes
14. **Can DAVID classify large gene lists into groups?** Correct Answer: Yes, into functionally related groups

15. **What type of studies provide datasets for DAVID?** Correct Answer: Genomic studies such as microarray

Microarray Data & Analysis

1. **What is a Microarray?** Correct Answer: A collection of microscopic DNA spots attached to a solid surface
2. **What is the main use of Microarrays?** Correct Answer: To measure expression levels of large numbers of genes simultaneously
3. **What are the two common types of color arrays?** Correct Answer: One colour and Two colour arrays
4. **Which process in microarray analysis involves identifying pixels of the feature?** Correct Answer: Feature extraction
5. **What is the purpose of Normalisation in microarray data?** Correct Answer: To remove systematic experimental errors and bias
6. **What does a HeatMap visualize?** Correct Answer: Gene expression data
7. **In a HeatMap, what does the color red usually represent?** Correct Answer: Up-regulated genes
8. **In a HeatMap, what does the color green usually represent?** Correct Answer: Down-regulated genes
9. **In a HeatMap, what does the color black represent?** Correct Answer: Unchanged expression
10. **What do the rows in a microarray data grid represent?** Correct Answer: Genes
11. **What do the columns in a microarray data grid represent?** Correct Answer: Samples
12. **What is Gene Set Enrichment Analysis (GSEA)?** Correct Answer: An approach to interpreting gene expression data based on functional annotation
13. **Which ontology is commonly used for gene set enrichment?** Correct Answer: Gene Ontology (GO)
14. **Name a popular tool for gene set enrichment.** Correct Answer: GSEA (or DAVID)
15. **What is a limitation of microarray data?** Correct Answer: Data variability for genes with low expression levels
16. **How can false positives be reduced in microarray experiments?** Correct Answer: Replication
17. **What is aRNA?** Correct Answer: Amplified RNA

18. **What does MIAME stand for?** Correct Answer: Minimal Information About a Microarray Experiment
19. **What is the underlying principle of microarray technology?** Correct Answer: The ability of DNA to bind to itself and to RNA
20. **Which type of array uses 25-nucleotide probes?** Correct Answer: Affymetrix
21. **What is photolithography used for in microarrays?** Correct Answer: Manufacturing Affymetrix arrays
22. **What is the purpose of a mismatch pair in Affymetrix?** Correct Answer: To detect and eliminate false results
23. **What is the typical file format for scanned microarray data?** Correct Answer: .gpr (GenePix Results)
24. **What is log2 transformation used for in microarray data?** Correct Answer: To treat up and down regulation symmetrically
25. **What does a log ratio of 0 mean?** Correct Answer: Not differentially expressed

Gene Networks & STRING

1. **What are the two basic components of a network?** Correct Answer: Nodes and Edges
2. **In a protein network, what do nodes represent?** Correct Answer: Proteins (or genes)
3. **In a network, what do edges represent?** Correct Answer: Interactions
4. **What is the “Degree” of a node?** Correct Answer: The number of edges that connect a node
5. **What does STRING stand for?** Correct Answer: Search Tool for the Retrieval of Interacting Genes/Proteins
6. **What is the primary function of STRING?** Correct Answer: A database of known and predicted protein-protein interactions
7. **What do interaction scores in STRING indicate?** Correct Answer: Indicators of confidence
8. **What is the range of scores in STRING?** Correct Answer: 0 to 1
9. **What does a score of 1 in STRING represent?** Correct Answer: Highest confidence
10. **How many types of evidence are used in STRING for predictions?** Correct Answer: Seven
11. **What does “Confidence Mode” in STRING show?** Correct Answer: Thickness of lines indicates the degree of confidence

12. **What does “Evidence Mode” in STRING show?** Correct Answer: Edges drawn with colored lines representing types of evidence
13. **What does “Action Mode” in STRING show?** Correct Answer: Additional info like binding or activation
14. **Can you search STRING by sequence?** Correct Answer: Yes
15. **What is the main entry point for STRING?** Correct Answer: The protein search box on the start page
16. **What does a red square in the STRING co-expression view represent?** Correct Answer: Higher association score of expression data
17. **What is a Gene Regulatory Network?** Correct Answer: A network that governs gene expression levels of mRNA and proteins
18. **What can be a regulator in a gene network?** Correct Answer: A gene, DNA, RNA, or Protein
19. **Are interactions in a network always direct?** Correct Answer: No, they can be direct or indirect
20. **What is morphogenesis?** Correct Answer: The creation of body structures

Cytoscape

1. **What is Cytoscape?** Correct Answer: An open-source software platform for visualizing and integrating molecular interaction networks
2. **How many ways are there to create networks in Cytoscape?** Correct Answer: Four
3. **Which menu in Cytoscape allows importing networks?** Correct Answer: File menu
4. **Can Cytoscape import data from Excel files?** Correct Answer: Yes
5. **What is the purpose of the “Layout” menu in Cytoscape?** Correct Answer: Visually organizing the network
6. **What is a “Style” in Cytoscape?** Correct Answer: A set of encoded table data sets (color, size, shape)
7. **Which plugin determines statistically overrepresented GO categories?** Correct Answer: BiNGO
8. **What does NetworkAnalyzer calculate?** Correct Answer: Topological parameters of the network
9. **What is “Betweenness Centrality”?** Correct Answer: A measure of a node’s centrality in a network
10. **Can Cytoscape create sub-networks?** Correct Answer: Yes

11. **What is the function of the “Apps” menu?** Correct Answer: Managing and accessing Cytoscape plugins
12. **How do you add a node manually in Cytoscape?** Correct Answer: Right-click on empty space and select Add -> Node
13. **What is a “Nested Network”?** Correct Answer: A network associated with a specific node
14. **What does the “Scale” tool in Cytoscape do?** Correct Answer: Changes the length of edges
15. **Which layout algorithm uses node or edge columns?** Correct Answer: Attribute-based layouts (e.g., Attribute Circle Layout)

Section 2: Python Programming

Python Basics

1. **Is Python a high-level or low-level language?** Correct Answer: High-level language
2. **What is an Interpreter?** Correct Answer: A program that processes high-level language into low-level language line by line
3. **What character is used for comments in Python?** Correct Answer: #
4. **How do you create a variable in Python?** Correct Answer: Using an assignment statement (e.g., x = 5)
5. **Does Python require variable types to be declared?** Correct Answer: No
6. **What is the output of 2**3 in Python?** Correct Answer: 8
7. **Which operator is used for the remainder of a division?** Correct Answer: % (Modulo)
8. **What is the order of operations in Python?** Correct Answer: PEMDAS
9. **What does the `print()` function do?** Correct Answer: Prints a value to the screen
10. **What is the result of `type("Hello")`?** Correct Answer:
11. **Which function converts a value to a string?** Correct Answer: `str()`
12. **Which function converts a value to an integer?** Correct Answer: `int()`
13. **What is the comparison operator for “equal to”?** Correct Answer: ==
14. **What is the comparison operator for “not equal to”?** Correct Answer: !=
15. **What is an `if` statement used for?** Correct Answer: Conditional execution

Strings & Loops

1. **Are strings in Python mutable?** Correct Answer: No, they are immutable
2. **How do you access the first character of string `s`?** Correct Answer: `s[0]`
3. **What does `s[1:4]` do?** Correct Answer: Slices the string from index 1 to 3
4. **Which method converts a string to lowercase?** Correct Answer: `lower()`
5. **Which method finds the position of a substring?** Correct Answer: `find()`
6. **Which method replaces a substring?** Correct Answer: `replace()`
7. **What does `s.count('A')` do?** Correct Answer: Counts occurrences of 'A' in string `s`
8. **What is a `for` loop used for?** Correct Answer: Iterating over a sequence
9. **What is a `while` loop?** Correct Answer: A loop that continues as long as a condition is true
10. **What does the `range(5)` function return?** Correct Answer: Numbers from 0 to 4
11. **What is the purpose of the `break` statement?** Correct Answer: To exit a loop prematurely
12. **What is the purpose of the `continue` statement?** Correct Answer: To skip the rest of the current iteration and move to the next
13. **How do you concatenate two strings?** Correct Answer: Using the + operator
14. **What does `s.upper()` do?** Correct Answer: Converts string to uppercase
15. **What is string indexing?** Correct Answer: Accessing characters by their position

Lists, Tuples, Dictionaries & Sets

1. **What is a List in Python?** Correct Answer: A mutable, ordered collection of items
2. **How do you add an item to a list?** Correct Answer: Using the `append()` method
3. **What does `list.pop()` do?** Correct Answer: Removes and returns the last item
4. **What is a Tuple?** Correct Answer: An immutable, ordered collection of items
5. **How do you define a single-item tuple?** Correct Answer: `(item,)` (with a comma)
6. **What is a Dictionary?** Correct Answer: A collection of key-value pairs
7. **Are dictionary keys unique?** Correct Answer: Yes
8. **Can a list be a dictionary key?** Correct Answer: No, because lists are mutable
9. **What does `dict.keys()` return?** Correct Answer: A list of all keys in the dictionary
10. **What is a Set?** Correct Answer: An unordered collection of unique items

11. **How do you add an element to a set?** Correct Answer: Using the `add()` method
12. **What is the intersection of two sets?** Correct Answer: Elements common to both sets
13. **What is the union of two sets?** Correct Answer: All unique elements from both sets
14. **Which list method sorts the list in place?** Correct Answer: `sort()`
15. **Which function returns a sorted copy of a list?** Correct Answer: `sorted()`

Modules, Classes & Exceptions

1. **What is a Module in Python?** Correct Answer: A file containing Python code (.py)
2. **How do you import a module?** Correct Answer: Using the `import` statement
3. **What is a Package?** Correct Answer: A directory containing a `__init__.py` file and modules
4. **What is a Class?** Correct Answer: A blueprint for creating objects (user-defined data types)
5. **What is an Object?** Correct Answer: An instance of a class
6. **What does `self` represent in a class method?** Correct Answer: The instance of the object itself
7. **What is a Constructor in Python?** Correct Answer: The `__init__` method
8. **What is Inheritance?** Correct Answer: A class taking attributes and methods from a superclass
9. **What is an Exception?** Correct Answer: An error that occurs during program execution
10. **Which block is used to handle exceptions?** Correct Answer: `try...except`
11. **What is a `ValueError`?** Correct Answer: An error when a function receives an argument of the right type but inappropriate value
12. **What is an `IOError`?** Correct Answer: An error related to input/output operations (e.g., file not found)
13. **What does `data.close()` do?** Correct Answer: Closes an open file object
14. **What is a private attribute in a class?** Correct Answer: An attribute starting with two underscores (e.g., `__x`)
15. **What is a docstring?** Correct Answer: A string used to document a module, class, or function

Section 3: Biopython & Phylogenetics

Biopython

1. **What is Biopython?** Correct Answer: A set of tools for computational molecular biology
2. **Which module is used for sequences in Biopython?** Correct Answer: `Bio.Seq`
3. **How do you get the complement of a sequence in Biopython?** Correct Answer: `my_seq.complement()`
4. **How do you get the reverse complement?** Correct Answer: `my_seq.reverse_complement()`
5. **Can Biopython parse FASTA files?** Correct Answer: Yes
6. **What does the `alphabet` attribute in `Seq` define?** Correct Answer: The type of sequence (DNA, RNA, or Protein)
7. **Which Biopython module is used for BLAST?** Correct Answer: `Bio.Blast`
8. **Can Biopython handle GenBank files?** Correct Answer: Yes
9. **What is the standard sequence class in Biopython?** Correct Answer: `Seq`
10. **Which tool is used for sequence alignment in Biopython?** Correct Answer: `Bio.Align` (or interfaces to ClustalW)

Phylogenetics

1. **What is Phylogenetics?** Correct Answer: The study of evolutionary relationships
2. **What is a Phylogenetic Tree?** Correct Answer: A representation of evolutionary relationships between organisms
3. **What do nodes in a tree represent?** Correct Answer: Taxonomic units
4. **What is a “Leaf” in a tree?** Correct Answer: A terminal unit at the tip of a branch (species or gene)
5. **What is a “Rooted Tree”?** Correct Answer: A tree with a common ancestor for all units
6. **What is an “Unrooted Tree”?** Correct Answer: A tree that specifies relationships but not an evolutionary path
7. **What is a Cladogram?** Correct Answer: A tree depicting only the branching order
8. **What is a Phylogram?** Correct Answer: A tree where branch lengths represent divergence
9. **What is an Ultrametric Tree?** Correct Answer: A rooted tree where all paths from root to leaf are equal

10. **What is the “Maximum Parsimony” method?** Correct Answer: A method that finds the tree with the fewest mutations
11. **What are “Informative Sites”?** Correct Answer: Positions in an alignment that support a specific topology
12. **What is the “Maximum Likelihood” method?** Correct Answer: A statistical method that finds the most likely tree given the data
13. **What is UPGMA?** Correct Answer: Unweighted Pair-Group Method using arithmetic Averages
14. **Does UPGMA assume a constant mutation rate?** Correct Answer: Yes (Molecular Clock Hypothesis)
15. **What is the Fitch-Margoliash method?** Correct Answer: A distance-based method that does not assume a constant mutation rate
16. **What is a “Distance Matrix”?** Correct Answer: A table showing pairwise distances between sequences
17. **What is the “E-value” in BLAST?** Correct Answer: The number of hits expected by chance (lower is better)
18. **What is a “Bootstrap” value?** Correct Answer: A measure of statistical support for a branch in a tree
19. **What is “Orthology”?** Correct Answer: Genes in different species that evolved from a common ancestral gene
20. **What is “Paralogy”?** Correct Answer: Genes related by duplication within a genome

Section 4: Statistics & Advanced Topics

Statistics in Bioinformatics

1. **What is a t-test?** Correct Answer: A statistical test to compare means of two groups
2. **What is a p-value?** Correct Answer: The probability of obtaining results as extreme as the observed ones by chance
3. **What is the common threshold for significance?** Correct Answer: $p < 0.05$
4. **What is a “Null Hypothesis”?** Correct Answer: The assumption that there is no difference or effect
5. **What is a “Type I Error”?** Correct Answer: A false positive (rejecting a true null hypothesis)

6. **What is a “Type II Error”?** Correct Answer: A false negative (failing to reject a false null hypothesis)
7. **What is the “Multiplicity Issue”?** Correct Answer: The problem of increased false positives when doing many tests
8. **What is “Bonferroni Correction”?** Correct Answer: Adjusting alpha by dividing it by the number of tests
9. **What is “False Discovery Rate” (FDR)?** Correct Answer: The expected proportion of false positives among significant results
10. **Which distribution is used for RNASeq data?** Correct Answer: Negative binomial distribution
11. **What is “Degrees of Freedom”?** Correct Answer: A parameter that controls the shape of the t-distribution
12. **What is a “One-sample t-test”?** Correct Answer: Testing a sample mean against a standard value
13. **What is an “Unpaired t-test”?** Correct Answer: Comparing means of two independent samples
14. **What is a “Paired t-test”?** Correct Answer: Comparing means of two correlated samples
15. **What is a “ROC Curve”?** Correct Answer: A plot showing sensitivity vs. specificity

Cancer Genomics & cBioPortal

1. **What is cBioPortal?** Correct Answer: An open-access resource for interactive exploration of cancer genomics data
2. **What kind of data does cBioPortal store?** Correct Answer: DNA copy-number, mRNA expression, mutations, etc.
3. **What is an “OncoPrint”?** Correct Answer: A concise graphical summary of genomic alterations across samples
4. **What does the “Mutual Exclusivity” tab show?** Correct Answer: Statistical analysis of co-occurrence of alterations
5. **Can cBioPortal perform survival analysis?** Correct Answer: Yes
6. **What is a “Mutation” in cancer genomics?** Correct Answer: A change in the DNA sequence of a cell
7. **What is “Copy Number Alteration” (CNA)?** Correct Answer: A change in the number of copies of a gene

8. **What is “DNA Methylation”?** Correct Answer: A chemical modification of DNA that can affect gene expression
9. **What is the “Network” tab in cBioPortal?** Correct Answer: Shows interactions between queried genes
10. **Is cBioPortal open-source?** Correct Answer: Yes

Additional Concepts

1. **What is “Transcriptomics”?** Correct Answer: The study of the transcriptome (all RNA transcripts)
2. **What is “Proteomics”?** Correct Answer: The study of the proteome (all proteins)
3. **What is “Functional Annotation”?** Correct Answer: Attaching biological information to gene sequences
4. **What is “Co-expression”?** Correct Answer: Genes that show similar expression patterns across conditions
5. **What is “Clustering”?** Correct Answer: Grouping objects based on similarity
6. **What is “Biological Interpretation”?** Correct Answer: Understanding the biological meaning of data results
7. **What is “Feature Extraction”?** Correct Answer: Transforming raw data (images) into measurements
8. **What is “Batch Analysis”?** Correct Answer: Performing analysis on multiple datasets at once
9. **What is “Topological Analysis”?** Correct Answer: Studying the structure and properties of a network
10. **What is a “Biological Pathway”?** Correct Answer: A series of actions among molecules in a cell that leads to a certain product or change
11. **What is “Synonymous Mutation”?** Correct Answer: A mutation that does not change the amino acid sequence
12. **What is “Non-synonymous Mutation”?** Correct Answer: A mutation that changes the amino acid sequence
13. **What is “HGT”?** Correct Answer: Horizontal Gene Transfer
14. **What is “Gene Duplication”?** Correct Answer: A major force in modifying genetic information
15. **What is “Gene Loss”?** Correct Answer: Deletion of genetic information

16. **What is a “Molecular Clock”?** Correct Answer: A technique that uses mutation rates to deduce the time of divergence
17. **What is “MSA”?** Correct Answer: Multiple Sequence Alignment
18. **What is “ClustalW”?** Correct Answer: A popular software for multiple sequence alignment
19. **What is “MEGA”?** Correct Answer: Molecular Evolutionary Genetics Analysis software
20. **What is “Bioconductor”?** Correct Answer: An open-source project for the analysis of genomic data using R